

Project Development Timeline (8 Weeks)

Week 1-2: Requirements & Data Engineering

Acquired baseline global HDI datasets. Handled missing records via country-specific medians. Log-transformed GNI indicators and normalized variables using StandardScaler.

Week 3-4: Machine Learning Pipeline

Benchmarked 5 algorithms. Trained and validated models. SVM emerged as the best classifier (98.9% F1-score). Serialized models into Joblib format.

Week 5-6: Web Application Frontend & Backend

Built lightweight Flask server. Designed the SPA layout with interactive sliders, Chart.js graphics, and dynamic theme switching.

Week 7-8: Advanced Features & Restructuring

Integrated circular gauge indicators, Did You Know facts, dynamic PDF downloads, and client-side CSV exports. Restructured to SDLC standards.